

Evidence Audit Companion to *Fermionic Extensions of Continuous Functional Tensor Networks*

Prepared for independent human review

Abstract

This companion reorganizes the evidence behind the associated PRA manuscript for line-by-line human review. It is not a second paper and introduces no additional scientific claim. For every major conclusion it states the input domain, the proof dependency, the reduction or calculation, the quantities that can be independently recomputed, and the boundary beyond which the conclusion does not extend. The final section gives a complete audit trail for the selected interacting calculation, including the state class, Hamiltonian, operator construction, optimization schedule, initialization lineage, reference calculation, error measures, resource measurements, and antisymmetry residual. Finite-size arithmetic checks can detect transcription errors, but only the displayed arguments and the cited source theorems support claims at arbitrary size.

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1 How to use this companion

The associated PRA manuscript is the publication text. This document is a review aid. A reviewer may audit the mathematical claims independently of the floating-point calculation and should record any unverified dependency rather than infer it from a successful small case. In particular, the two reductions that use Chien–Harsha–Sinclair–Srinivasan (CHSS) [1] require independent checking of the cited source theorem, its field assumptions, and the row-order convention.

The following boundaries apply throughout.

1. Exact worst-case hardness does not exclude Monte Carlo, additive approximation, or an efficiently contractible promised subclass.
2. The ordinary particle-TT rank statements do not apply automatically to occupation-number MPS, ordered-coordinate methods, or exterior cores.
3. A floating-point result is numerical evidence, not a theorem.
4. The selected numerical state is a finite nonorthogonal Slater expansion and is therefore not claimed to be distinct from NOCI.
5. Antisymmetry is exact in the exterior representation; every numerical summary nevertheless reports the corresponding residual.

2 Claim map and dependency structure

Item	Claim	Principal support	Independent dependency
E1	Exact particle-TT bonds equal particle-cut matrix ranks.	Successive rank factorizations (TT-SVD).	Oseledets [2].
E2	Every nonzero alternating N -tensor has cut rank at least $\binom{N}{k}$.	Explicit diagonal minor selected from one nonzero Pluecker coordinate.	None.
E3	A Slater determinant has a flat particle Schmidt spectrum and the stated optimal rank- r error.	Partition of the permutation sum plus Eckart–Young.	Coleman [3].
E4	Exact squared norm is #P-hard for rational one-form FEMPS at maximum bond two.	Direct Cayley-coefficient lemma and one-query metric reduction.	CHSS Theorems 3.5 and 3.9.
E5	Exact point evaluation is #P-hard in an explicit rational shifted- Legendre basis at maximum bond two.	CHSS input, invertible evaluation matrix, explicit bit bounds, and one-query postprocessing.	CHSS plus standard Vandermonde identities.
E6	Specified matrix-pair memories collapse to polynomial LC–AGP families.	Radical filtration or fixed-counter interpolation followed by polarization.	Supplied algebra decomposition or embedding is part of the input.

Item	Claim	Principal support	Independent dependency
E7	A bandwidth-one pair path encodes a permanent; real-PSD relative norm inherits the stated obstruction.	Unique path and even-form commutativity.	Valiant [4]; Meiburg [5].
E8	A universal direct statistics-carrier tensor product fails.	Explicit two-Slater form with one-cut rank $N + 2$.	None.
E9	The selected $N = 6, D = 12, K = 4$ interacting calculation has the reported energy, error, variance, norm, cost, and exact antisymmetry.	Determinant transition solver and independent same-basis exterior CI.	Numerical recomputation; no theorem inference.

3 Common definitions and normalization

Let V_D be a D -dimensional real or complex one-particle space with orthonormal basis $e_1, \dots, e_D$. We use the isometric alternating embedding

$$J_N(v_1 \wedge \dots \wedge v_N) = \frac{1}{\sqrt{N!}} \sum_{\pi \in S_N} \text{sgn}(\pi) v_{\pi(1)} \otimes \dots \otimes v_{\pi(N)}. \quad (1)$$

A one-form matrix-wedge state has cores $A^{[j]} \in \text{Mat}_{\chi_{j-1} \times \chi_j}(V_D)$, $\chi_0 = \chi_N = 1$, and

$$C(A) = [A^{[1]} \wedge \dots \wedge A^{[N]}]_{11} \in \Lambda^N V_D. \quad (2)$$

Every virtual path is a wedge of N one-forms. Thus Eq. (2) is alternating without an approximate projection. The structural antisymmetry residual is therefore exactly zero. Whenever a labelled-particle tensor is materialized for a small system, an independent residual is

$$\epsilon_{\text{AS}}^{\text{mat}} = \max_{1 \leq i < N} \frac{\|C + \tau_{i,i+1} C\|_2}{\|C\|_2}, \quad (3)$$

where $\tau_{i,i+1}$ swaps adjacent particle slots.

Reviewer decision: ☐ verified ☐ revision needed ☐ outside expertise

4 E1–E3: exchange rank in particle factorization

4.1 E1: exact particle-TT rank

Theorem 4.1. *For every coefficient tensor C and cut $k \mid N - k$, the smallest exact ordinary particle-TT bond is*

$$r_k^{\text{TT}}(C) = \text{rank } C_{(k)}. \quad (4)$$

Proof. Contract the first k cores of any exact TT into a matrix L_k and the remaining cores into R_k . Then $C_{(k)} = L_k R_k$, so the bond is at least $\text{rank } C_{(k)}$. Conversely, factor the first unfolding at its exact rank, reshape the right factor, and repeat. At the k th step the carried index has dimension exactly $\text{rank } C_{(k)}$. The successive factorizations reconstruct C , establishing simultaneous attainability at all cuts. $\square$

Reviewer question. Does the proof establish simultaneous TT ranks, rather than unrelated factorizations at individual cuts? Is the theorem restricted to the labelled-particle tensor product?

Reviewer decision: ☐ verified ☐ revision needed ☐ outside expertise

4.2 E2: universal exchange floor

Theorem 4.2. *If $0 \neq C \in \Lambda^N V_D$, then $\text{rank } C_{(k)} \geq \binom{N}{k}$ for $1 \leq k < N$. Equality is attained by every nonzero decomposable N -form.*

Proof. Choose an N -element set I whose alternating coefficient c_I is nonzero. Restrict the row index of $C_{(k)}$ to the k -subsets $A \subset I$ and the column index to their complements $I \setminus A$. The resulting $\binom{N}{k} \times \binom{N}{k}$ submatrix is diagonal up to row and column signs: the matched entry is $\pm c_I$, while a mismatched row and column repeat at least one basis index and hence have zero alternating coefficient. Its determinant is nonzero, proving the lower bound. If $C = u_1 \wedge \cdots \wedge u_N$, all contractions lie in $\Lambda^{N-k} \text{span}\{u_1, \dots, u_N\}$, which has dimension $\binom{N}{k}$; the preceding lower bound is therefore sharp. $\square$

The immediate truncation consequence is that a nonzero ordinary particle TT with a smaller bond at any cut cannot remain exactly alternating. This does not say that a mode-space MPS needs the same bond.

Reviewer decision: ☐ verified ☐ revision needed ☐ outside expertise

4.3 E3: flat Slater spectrum

Theorem 4.3. *For orthonormal $u_1, \dots, u_N$, the normalized Slater state Φ has*

$$\Phi = \binom{N}{k}^{-1/2} \sum_{|I|=k} (-1)^{\epsilon(I)} \Phi_I \otimes \Phi_{I^c}. \quad (5)$$

All $\binom{N}{k}$ singular values are $\binom{N}{k}^{-1/2}$, and the best cut-rank r approximation has relative squared error $1 - r/\binom{N}{k}$.

Proof. Partition the $N!$ signed permutations in Eq. (1) by the set I of orbitals appearing in the first k slots. Each class contains $k!(N-k)!$ terms and factors into normalized Slater states Φ_I and Φ_{I^c} with coefficient $\sqrt{k!(N-k)!/N!} = \binom{N}{k}^{-1/2}$. Different subsets are orthogonal on both sides. Equation (5) is therefore a Schmidt decomposition. Discarding the smallest singular values is optimal by the Eckart–Young theorem; because they are all equal, retaining r leaves the stated squared error. $\square$

Reviewer question. Confirm that the orbitals are orthonormal and that no flat-spectrum statement is made for a general correlated alternating tensor.

Reviewer decision: ☐ verified ☐ revision needed ☐ outside expertise

5 E4: fixed-bond exact squared-norm obstruction

For a matrix $M = (M_{ij})$ over a noncommutative algebra, define the row-ordered Cayley determinant

$$\text{CDet}(M) = \sum_{\sigma \in S_N} \text{sgn}(\sigma) M_{1\sigma(1)} \cdots M_{N\sigma(N)}. \quad (6)$$

The product order is increasing in the row label. This is the essential difference from the ordinary scalar determinant.

Lemma 5.1 (direct coefficient). *Set $\mathcal{A}^{[i]} = \sum_j M_{ij}e_j$. Then for boundary vectors u, v ,*

$$u^\top \mathcal{A}^{[1]} \wedge \cdots \wedge \mathcal{A}^{[N]} v = (u^\top \text{CDet}(M)v) e_1 \wedge \cdots \wedge e_N. \quad (7)$$

Proof. A top-form term must select each e_j exactly once, hence is indexed by a permutation. Reordering its exterior factors contributes the permutation sign, while matrix multiplication retains row order. Summation gives Eq. (7). $\square$

Theorem 5.2. *Over $\mathbb{Q}$, exact squared-norm evaluation for rational one-form FEMPS is $\#P$ -hard under a polynomial-time metric reduction even when $D = N$ and all internal bonds are at most two.*

Proof. For a 3CNF formula φ with m clauses and S satisfying assignments, CHSS Theorems 3.5 and 3.9 construct in polynomial time a polynomial-size matrix H_φ over $\text{Mat}_2(\mathbb{Q})$ with constant-bit integer gadget weights and

$$\text{CDet}(H_\varphi) = aI_2 + bJ_2, \quad a + b = 4^{3m}S, \quad J_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (8)$$

Use Eq. (7) with $u = e_1$ and $v = e_1 + e_2$. The state has only one top-form coefficient, $u^\top(aI_2 + bJ_2)v = a + b = 4^{3m}S \geq 0$, and therefore norm $16^{3m}S^2$. A single exact norm query, exact nonnegative integer square root, and division by 4^{3m} recover S . The CHSS matrix size and core encoding are polynomial in $|\varphi|$; the queried integer has polynomial bit length because $S \leq 2^{n_{\text{var}}}$. Both the construction and postprocessing are polynomial-time, and the maximum bond stays two. $\square$

External dependency to check. The proof uses the structured CHSS output in Eq. (8), not merely hardness of an arbitrary signed matrix entry. The reviewer should confirm: (i) CHSS uses the same row-ordered convention; (ii) the construction is valid over characteristic zero; (iii) its gadget weights and graph size have polynomial encoding; and (iv) the stated boundary indeed selects $a + b$. For a generic signed Cayley entry the manuscript retains only the weaker bond-three polarization observation.

Reviewer decision: $\square$ verified $\square$ revision needed $\square$ outside expertise

6 E5: exact point evaluation in a rational Legendre basis

Let $\ell_r(t) = P_r(2t - 1)$ be the unnormalized shifted Legendre polynomial, $P_r(1) = 1$. Its integer expansion is

$$\ell_r(t) = \sum_{k=0}^r (-1)^{r-k} \binom{r}{k} \binom{r+k}{k} t^k, \quad (9)$$

with leading coefficient $\binom{2r}{r}$.

Theorem 6.1. *Exact point evaluation of rational one-form FEMPS in $\ell_0, \dots, \ell_{D-1}$ is $\#P$ -hard under a polynomial-time metric reduction already at $D = N$ and maximum bond two. Under the isometry Eq. (1), the output is represented exactly as $q/\sqrt{N!}$ with rational q .*

Proof. Choose rational nodes $t_i = i/(N + 1)$ and set $B_{ij} = \ell_{j-1}(t_i)$. Degree grading changes the monomial Vandermonde matrix by an upper-triangular coefficient matrix, so

$$\det B = \left(\prod_{r=0}^{N-1} \binom{2r}{r} \right) \prod_{1 \leq i < k \leq N} (t_k - t_i) \neq 0. \quad (10)$$

The nodes have $O(\log N)$ bits. Equation (9) gives polynomial-bit evaluations. More explicitly, $Q = (N + 1)^{N-1}$ clears all denominators in B . The integer entries of QB have polynomial bit length; Hadamard bounds applied to its determinant and minors imply polynomial bit length for

$$B^{-1} = Q \operatorname{adj}(QB) / \det(QB). \quad (11)$$

Thus the basis evaluation map and its inverse are exactly computable in polynomial Turing time.

Take the CHSS matrix H_φ from E4 and define

$$\mathcal{A}^{[i]}(t) = \sum_{j=1}^N H_{ij} \ell_{j-1}(t), \quad u = e_1, \quad v = e_1 + e_2. \quad (12)$$

In the alternating point expansion at $(t_1, \dots, t_N)$, repeated basis columns cancel. Each column permutation contributes its sign times $\det B$ and leaves the CHSS matrix product in row order. Hence

$$\Psi(t_1, \dots, t_N) = \frac{\det B}{\sqrt{N!}}, 4^{3m} \# \text{SAT}(\varphi). \quad (13)$$

The exact algebraic output is the pair (q, N) representing $q/\sqrt{N!}$; $N!$ has $O(N \log N)$ bits. One query followed by rational division by the known nonzero factor $4^{3m} \det B$ recovers $\# \text{SAT}(\varphi)$. $\square$

Separation from E4. E4 is a norm query on a coefficient state; E5 is a point query in a specified functional basis. They are separate reductions from the same CHSS source. Neither implies that all $\chi \geq 2$ instances are unsuitable for QMC or VMC.

Reviewer decision: ☐ verified ☐ revision needed ☐ outside expertise

7 E6–E7: tractable corridors and a sparse hard path

7.1 Bounded coefficient algebra

Let $\mathcal{A} = S \oplus R$ be a finite-dimensional complex algebra with $R^d = 0$ and $S = \bigoplus_{a=1}^q \operatorname{Mat}_{p_a}(\mathbb{C})$; put $p = \max p_a$ and $\rho = \dim R$. Expanding Ω^M by the number k of radical factors leaves only $k < d$. Resolving the $k + 1$ semisimple runs into Wedderburn blocks and the radical insertions in a basis gives at most $q^{k+1} \rho^k$ structural choices. For each choice, even physical two-forms commute, and after the boundary functional is applied one obtains a homogeneous degree- M polynomial in at most

$$v_k = (k + 1)p^2 + k \quad (14)$$

commuting two-forms. Degree- M powers span this polynomial space in characteristic zero. Therefore

$$K \leq \sum_{k=0}^{\min(d-1, M)} q^{k+1} \rho^k \binom{M + v_k - 1}{v_k - 1} \quad (15)$$

AGP terms suffice. For the rational Turing statement, the displayed Wedderburn–Malcev decomposition and radical basis must be supplied; the theorem does not claim an algorithm that discovers them.

7.2 Fixed-state graded memory

If the coefficient algebra is supplied with an embedding in

$$\operatorname{Mat}_w \left(\mathbb{C}[z_1, \dots, z_g] / (z_1^{d_1}, \dots, z_g^{d_g}) \right) \quad (16)$$

for fixed w, g , lift the quotient entries and note $\deg_{z_j} \Omega^M \leq M(d_j - 1)$. Tensor-product Vandermonde interpolation extracts the counter coefficients. At each grid point the boundary of the matrix power is a homogeneous polynomial in at most w^2 commuting two-forms, so polarization yields

$$K \leq \prod_{j=1}^g [M(d_j - 1) + 1] \binom{M + w^2 - 1}{w^2 - 1}. \quad (17)$$

Rational nodes, Hadamard bounds, and exact Gaussian elimination give the polynomial bit bound when the embedding is part of the input.

7.3 Sparse APG permanent path

For orthonormal pair forms $P_j = e_{2j-1} \wedge e_{2j}$ and a zero-one matrix A , set $F_i = \sum_j A_{ij} P_j$. The unique endpoint path of an upper-bidiagonal pair matrix produces

$$\Psi_A = \frac{F_1 \wedge \cdots \wedge F_M}{M!} = \frac{\text{perm}(A)}{M!} P_1 \wedge \cdots \wedge P_M. \quad (18)$$

Repeated P_j vanish, and distinct even pair forms commute, so precisely the permanent terms survive. The squared norm is $\text{perm}(A)^2 / (M!)^2$. Exact nonnegative square root recovers the zero-one permanent. For real positive-semidefinite rational A , a relative squared-norm approximation would similarly give a relative approximation to $\text{perm}(A)$; the manuscript uses Meiburg's result for the corresponding complexity consequence. Entrywise-nonnegative permanents have an FPRAS [6], so the promises must not be conflated.

Reviewer decision: ☐ verified ☐ revision needed ☐ outside expertise

8 E8: failure of a universal direct statistics carrier

For $C \in \Lambda^N V$, consider the one-cut contraction map $c_{C,1} : V^* \rightarrow \Lambda^{N-1} V$. A universal direct factorization with a Slater-normalized statistics carrier would force every admitted one-cut rank to be divisible by N , because a Slater has rank N and multiplicity one.

For $N \geq 3$ define in $V = \mathbb{F}^{N+2}$

$$C_N = e_1 \wedge (e_2 \wedge e_3 + e_4 \wedge e_5) \wedge e_6 \wedge \cdots \wedge e_{N+2}. \quad (19)$$

This is a sum of two Slaters. Contracting successively by the duals of $e_1, \dots, e_{N+2}$ gives $N+2$ forms with distinct monomial supports, hence linearly independent images. Therefore $\text{rank } c_{C_N,1} = N+2$, while $N \nmid N+2$. The proposed universal direct tensor product cannot hold for all nonzero N -forms. The proof does not exclude state-adaptive factorizations, Hamiltonian-specific charges, or a different categorical carrier with its own reconstruction theorem.

Reviewer decision: ☐ verified ☐ revision needed ☐ outside expertise

9 E9: selected interacting numerical calculation

9.1 State class and physical model

The numerical state is the nonbranching diagonal-path subclass

$$\Psi = \sum_{a=1}^K c_a \Phi_a, \quad \Phi_a = u_{a1} \wedge \cdots \wedge u_{aN}. \quad (20)$$

Equation (20) is exactly a K -term nonorthogonal Slater expansion. It is a restricted FEMPS realization, but not a state class beyond NOCI. The selected calculation uses $N = 6$, $D = 12$, and $K = 4$ in the first twelve one-dimensional harmonic-oscillator functions. The Hamiltonian is

$$H = \sum_{i=1}^N \left(-\frac{1}{2} \partial_{x_i}^2 + \frac{1}{2} x_i^2 \right) + \sum_{i < j} \frac{1}{\sqrt{(x_i - x_j)^2 + 1}}. \quad (21)$$

The soft-Coulomb tensor is projected with 128-point Gauss–Hermite quadrature. A 160-point calculation changes that tensor relatively by 1.3545×10^{-12} . A physical-index SVD retains 23 factors at $D = 12$; the relative reconstruction error is 3.8469×10^{-15} . The independent reference uses the unfactorized four-index quadrature tensor and direct Slater–Condon diagonalization in the $\binom{12}{6} = 924$ dimensional exterior space.

9.2 Determinant transition algorithm

Let $U_a \in \mathbb{C}^{D \times N}$ collect the orbitals in Φ_a . For each of the K^2 ordered transition pairs,

$$S_{ab} = U_a^\dagger U_b, \quad \langle \Phi_a | \Phi_b \rangle = \det S_{ab}. \quad (22)$$

For a one-body matrix h , with $X_{ab} = U_a^\dagger h U_b$,

$$\langle \Phi_a | H_1 | \Phi_b \rangle = \det(S_{ab}) \operatorname{Tr}(S_{ab}^{-1} X_{ab}). \quad (23)$$

For a factorized two-body term $L_\ell \otimes R_\ell$, set $A_{ab}^\ell = U_a^\dagger L_\ell U_b$ and $B_{ab}^\ell = U_a^\dagger R_\ell U_b$. Its transition value is

$$\det(S_{ab}) \left[\operatorname{Tr}(S_{ab}^{-1} A_{ab}^\ell) \operatorname{Tr}(S_{ab}^{-1} B_{ab}^\ell) - \operatorname{Tr}(S_{ab}^{-1} A_{ab}^\ell S_{ab}^{-1} B_{ab}^\ell) \right]. \quad (24)$$

Near a singular overlap, the implementation uses column-replacement cofactors instead of an inverse. In the selected run all 16 transition pairs passed the conditioning criterion and used the inverse route. No virtual path history and no D^N labelled-particle coefficient tensor were enumerated.

9.3 Optimization and initialization lineage

Each U_a is returned to a Stiefel gauge by reduced QR with deterministic diagonal phases. At every orbital step the amplitudes are eliminated by the conditioned generalized Hermitian eigenproblem

$$H(U)c = E S(U)c. \quad (25)$$

All four overlap directions were retained in the selected run; the final condition number was 2.1111 and the generalized residual norm was 1.96×10^{-14} . Automatic differentiation propagates through QR, transition contractions, and the generalized eigenvalue.

The nonlinear schedule used 160 Adam steps with a cosine learning-rate change from 10^{-3} to 10^{-5} , followed by 80 requested L-BFGS refinement steps at learning rate 0.5. The random seed was 3212. The starting orbitals were not a fresh $D = 12$ Slater and were not derived from CI. They were obtained by exact zero-padding of an already optimized $D = 10, K = 4$ state. The energy moved from 25.0498252875 at the start to 25.0494882497 after Adam and to 25.0494711446 after L-BFGS. This lineage must remain visible because it precludes a claim of blind-from-scratch $D = 12$ optimization.

9.4 Definitions of the reported checks

For the normalized exterior coefficient vector c_{ext} and the dense same-basis Hamiltonian H_{CI} ,

$$n = \langle c_{\text{ext}}, c_{\text{ext}} \rangle, \quad (26)$$

$$E = \langle c_{\text{ext}}, H_{\text{CI}} c_{\text{ext}} \rangle / n, \quad (27)$$

$$\sigma_E^2 = \|(H_{\text{CI}} - E)c_{\text{ext}}\|_2^2 / n. \quad (28)$$

The energy error is $E - E_{\text{CI}}$, where E_{CI} is the lowest eigenvalue of the same 924-dimensional dense exterior Hamiltonian. The norm error is $|n - 1|$. Structural antisymmetry is zero by Eq. (20); a full 12^6 particle tensor was deliberately not built, so no materialized residual is substituted for the structural result.

9.5 Results and resource account

Quantity	Value
Variational energy	25.0494711446
Same-basis CI energy	25.0493664161
Error relative to CI	1.0473×10^{-4}
Energy variance	1.1223×10^{-3}
Norm error	1.55×10^{-15}
Structural antisymmetry residual	0
Factorized/dense energy difference	2.84×10^{-14}
Wall time	107.7 s on CPU
Peak sampled process RSS	974,974,976 bytes
Exterior dimension	924
Transition pairs	16
Stored orbital and amplitude scalars	292
Enumerated virtual paths	0
Materialized particle coefficients	0

Table 2: Selected numerical result. The peak RSS is a sampled whole-process measurement, not an asymptotic memory bound.

The absolute FEMPS/NOCI and direct-CI energies both decrease along the tested $D = 8, 10, 12$ basis sequence, but the error relative to same-basis CI is not monotone at fixed $K = 4$. Consequently Table 2 is a finite-basis numerical illustration, not evidence for accelerated D convergence. It establishes neither a new ansatz beyond NOCI nor superiority over Li–Waintal, DMRG, particle TT, or FCI.

Reviewer question. Recompute the dense exterior norm, Rayleigh quotient, residual variance, and lowest CI eigenvalue from the final 924 coefficients. Confirm that the factorized and unfactorized Hamiltonian energies agree within the stated tolerance and that the zero-padded initialization is disclosed in any summary of the result.

Reviewer decision: ☐ verified ☐ revision needed ☐ outside expertise

10 Final human sign-off

The following sign-off is intentionally not prefilled.

Reviewer name and
affiliation:

Area of expertise:

Items reviewed (E1–E9):

Unresolved objections:

Required revisions:

Overall recommendation: ☐ accept evidence as stated ☐ revise ☐ reject

Signature/date:

References

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